

Production lines — API specification

005_DD-API version 2 supports SCR-005, FN-032–FN-036 and REQ-064–REQ-069. WI-009 approved plan revision 1, step 7; contracts only, not implemented.

1. Document control and references

Field	Value
Document ID / version	005_DD-API / 2
System / subsystem	ProductionManagementAI / Master data
Work item / author	WI-009 / Agent
Created / updated	2026-10-01 / 2026-10-01
State	Submitted for review; DEC-010 resolved; not implemented

Version	Date	Author	Change
1	2026-10-01	Agent	Initial API contracts, errors, concurrency and telemetry
2	2026-10-01	Agent	Finalize Admin/Operator read authorization after DEC-010 confirmation

Inputs: approved [005 BD](#), [005 DB](#) and [main 005 DD](#). [Requirements](#), [decisions](#) and [plan](#) define scope and review stops. No approved design is edited. FN/SPD and the new order-impact DD are reserved later documents, not existing links or completed specifications.

2. Transport, authorization and common bounds

Concern	Contract
Authentication	Existing same-origin Identity cookie; no new CORS/provider/role. Missing session returns established empty-body 401, not HTML redirect
Read policy	Admin or Operator for all seven API-PL operations, as confirmed in DEC-010; reuse ProductionOrderEditor policy, matching existing Product master access. Other authenticated roles receive 403; no new role or broader read entitlement
Write policy	Admin or Operator, server checked; UI visibility is not authority. Empty-body 403 retains existing convention
Representation	UTF-8 application/json; application/problem+json for application failures; no-store for successful reads, writes and application errors

Concern	Contract
Mutation	JSON object only, application/json (optional charset=utf-8); reject other media types with 415; no form/query mutation
Identity	Route/query/body UUID strings, nonzero; malformed UUID 400, valid unknown target 404; bind without a guid route constraint so malformed targets do not become accidental 404
Strict fields	New feature requests reject unknown/duplicate JSON member names, wrong scalar types, null required fields and unsupported actions with 400; existing API contracts are not globally tightened
Text	Trim both ends; Unicode code-point count after normalization. No extra code alphabet restriction. DB lower(code) comparison governs duplicates in all states
Request limits	New mutation body <=256 KiB; changes <=1000 per request; these are technical request limits, not a cap on persisted associations. Oversize 413; invalid action count 400
Query limits	q optional, trim, <=100 code points; page ASCII integer 1–10000, default 1; pageSize fixed 50; unknown/repeated query keys 400
List ordering	Line code ASC then UUID ASC, using DB collation; no user-defined SQL sort identifiers

Exact numeric and version representation

New line APIs send hours and minutes as JSON strings. Request decimal syntax is `^(0|[1-9][0-9]*)(\.[0-9]{1,3})?;$`; no whitespace, signs, exponent, grouping, NaN/infinity or rounding. Hours: $0 < \text{value} \leq 24$. Minutes/unit: $0 < \text{value} \leq 999999999.999$. Server parses invariant .NET decimal after lexical validation. Response is canonical non-exponent decimal text with unnecessary trailing zeros removed, e.g. "7.5", "0.001". New contracts use no JSON floating-point numbers for these fields; existing order quantity JSON-number contracts stay unchanged.

Line version is a canonical unsigned-decimal string for parent xmin (0–4294967295), not an incrementable client counter. Product unitRevision is a canonical string for bigint 0–9223372036854775807. Reject leading zeros except "0", numeric tokens and out-of-range values. Both are opaque to UI comparisons except exact equality. A unit revision is not the Product master edit version. Existing product/order numeric versions retain their current contract.

3. Operation catalog and registry

All proposed handlers belong to ProductionLinesController, except the existing order controller. Each has Agent / 2026-10-01 as creation/update metadata. API IDs are local registry entries, not duplicated IDs for existing order routes.

ID	Method / route	Handler / Application responsibility	Return
API-PL-01	GET /api/production-lines	List / FN-032	LinePage, 200
API-PL-02	GET /api/production-lines/{id}	Get / FN-033/034	LineDetail, 200
API-PL-03	POST /api/production-lines	Create / FN-033/034	LineDetail, 201

ID	Method / route	Handler / Application responsibility	Return
API-PL-04	PUT /api/production-lines/{id}	Update / FN-033/034	LineDetail, 200
API-PL-05	POST /api/production-lines/{id}/retire	Retire / FN-035	LineSummary, 200
API-PL-06	GET /api/production-lines/product-choices	ProductChoices / FN-034	ProductChoicePage, 200
API-PL-07	GET /api/production-lines/eligible	Eligible / FN-036	EligibleLinePage, 200
API-PO-01-04	Existing production-order create/get/update/list	Existing controller, FN-036 extension	Existing response plus nullable line

Literal choice/eligible routes precede the parameter route in API descriptions; route resolution tests must verify that they are not treated as line IDs. Full Application algorithms and client processing live in later FN/SPD documents.

Shared handler boundary

Step	Responsibility	Calls / owner
1	Authenticate/authorize, validate route/query/media/body and exact scalar syntax	Existing session policy; API validation
2	Call the scoped use case with cancellation/deadline	FN-032–FN-036 in later 005_DD-FN
3	Map known validation/conflict/not-found/transient outcome, no SQL detail	This API's error catalog
4	Return committed result/read projection and bounded operation telemetry	Representation below; FN instrumentation

4. Response field catalog

Fields are required unless explicitly nullable. Do not invent missing rows or units after integrity errors. UTC timestamps are ISO-8601 strings with Z.

Representation / field	JSON type / bound	Source / value mapping
LineSummary.id	UUID string	production_lines.id
LineSummary.code / name	string <=50 / <=200 code points	Stored immutable code / current name
LineSummary.workingHoursPerDay	decimal string	Exact hours/day, visible UI unit 「時間」 (hours)
LineSummary.isActive	boolean	true/false -> 「使用中」 / 「使用停止」 (Active / Retired)
LineSummary.updatedAt / version	UTC string / opaque string <=10	updated_at_utc / parent xmin
LinePage.items / total / page / pageSize	array <=50 / integer / integer / 50	Filtered count and requested page; count must be JS-safe integer

Representation / field	JSON type / bound	Source / value mapping
LineDetail	LineSummary fields plus pairs	Parent and page read from one snapshot
LineDetail.pairs	{items, total, page, pageSize}	All active/retired pairs; fixed 50, sorted product SKU then UUID; fields as PairDetail
PairDetail.product	{id, sku, name, unit, unitRevision, isActive}	Current product projection; sku/name <=50/200; unit one of seven; revision opaque <=19
PairDetail.minutesPerUnit	decimal string	Current coefficient; no history recomputation
PairDetail.confirmedUnit / confirmedUnitRevision	unit string / opaque string <=19	Persisted confirmation values
PairDetail.isActive / requiresUnitConfirmation	boolean / boolean	Pair state; mismatch of confirmed/current unit OR revision
PairDetail.updatedAt	UTC string	Pair update audit time; no independent editor version
ProductChoicePage	{items, total, page, pageSize}	Paged ProductChoice array, same current product fields as above
EligibleLinePage	{product, items, total, page, pageSize}	Product observation plus eligible summaries; items add minutesPerUnit and unit

Unit set is exactly 個, 本, 枚, 台, セット, kg, m. Unit mismatches remain readable even when an active pair is ineligible. No daily quantity, capacity total, scheduling estimate, historical duration or credential field.

5. Endpoint contracts

5.1 API-PL-01 — List

Arguments: query q, state, page. state is active (default), retired or all; unknown/case-variant value is 400. q searches literal code/name substrings, case-insensitive under DB comparison; escape LIKE wildcard characters rather than treating user text as SQL/pattern syntax. No extra sort/pageSize parameter. 200 LinePage; page beyond available results returns empty items with actual total. Read count and rows in one read-only REPEATABLE READ snapshot. No lock reservation. Errors: 400, 401/403, 503 or generic 500; no fabricated fallback page.

5.2 API-PL-02 — Detail

Arguments: route id, query pairsPage (default 1, 1–10000). 200 LineDetail; 404 if target absent. Include retired lines/pairs. Read parent, pair count/page and current products in one read-only REPEATABLE READ snapshot to avoid mixed parent/pair versions. Pair pagination does not mutate or omit history from storage. Client loading another page compares returned parent version with its baseline; a changed version requires a guarded reload, not a silent merge. Product unit changes can occur without changing parent xmin, so every timing intent also carries its observed unit revision. Errors: 400, 401/403, 404, 503, 500.

5.3 API-PL-03 — Create

No route/query arguments. Body CreateLineRequest:

Field	Type / required	Bound / meaning / example
code	string / yes	Trim 1–50 code points; "LINE-A"; unique including retired lines
name	string / yes	Trim 1–200 code points; Japanese permitted
workingHoursPerDay	decimal string / yes	"8", exact approved hour bound
products	array / yes, may be empty	<= 1000 AddProductInput rows; no duplicate productId
products[i].productId	UUID string / yes	Existing active product; server verifies under lock
products[i].minutesPerUnit	decimal string / yes	"0.125", exact approved coefficient bound
products[i].expectedUnit	unit string / yes	Unit displayed to user
products[i].expectedUnitRevision	opaque string / yes	Revision displayed with that unit
products[i].confirmUnit	boolean true / yes	Explicit confirmation of coefficient for observed unit

Server generates line ID/state/timestamps/version; body cannot set them. Save line and all pairs atomically. Locked product must still match observed unit and revision; do not silently confirm an unseen newer unit. Zero products is valid. 201 Location: /api/production-lines/{id}; LineDetail (pairs page 1). Read back the committed projection; failure after commit is not proof of rollback. Errors: 400 structural/product validation, 409 duplicate code/unit race, 401/403, 413/415, 503 known rolled-back transient failure, 500 unexpected result.

5.4 API-PL-04 — Update aggregate

Arguments: route id; no query. Body UpdateLineRequest:

Field	Type / required	Rule
name	string / yes	Same trim/bound as create
workingHoursPerDay	decimal string / yes	Same hour bound
version	opaque string / yes	Last read parent version; stale gives 409
productChanges	array / yes, may be empty	<= 1000 actions, one per productId; omitted pairs unchanged, never deleted
productChanges[i].action	string / yes	Exactly add, setTiming or retire

Field	Type / required	Rule
productChanges[i].productId	UUID string / yes	Immutable saved identity or active product to add

Action-specific fields (all other fields rejected):

action	Additional fields	Semantics
add	minutesPerUnit, expectedUnit, expectedUnitRevision, confirmUnit=true	Same rules as AddProductInput; saved retired pair cannot be re-added
setTiming	minutesPerUnit, expectedUnit, expectedUnitRevision, confirmUnit boolean	Saved active pair; observed unit/revision must match current product. If saved confirmation is stale, require true; false may edit current-unit timing without replacing confirmation
retire	None	Saved active pair becomes inactive; rows/orders retained. No timing edit or restore bundled into this action

code, id, isActive, timestamps and confirmed fields are not mutable payload members. Retired-line name/hours corrections retain inactive state. Retired pair is read-only. New association requires active product; existing timing edits do not reactivate inactive products. Plain name/hour edits may leave stale pairs unchanged. Parent/pair operations form one transaction; any failure rolls back all requested changes. Parent xmin must change even for pair-only mutations. 200 LineDetail (page 1), new parent version. Errors: 400 malformed/immutable/ duplicate actions/invalid product/pair operation; 404 line/pair absent; 409 stale parent or unit, confirmation required; 401/403, 413/415, 503, 500. Compare parent version before reporting business conflicts, after globally ordered lock acquisition.

5.5 API-PL-05 — Retire line

Arguments: route id; body only {"version":"81234"}. Current active line with matching parent version becomes inactive. No implicit pair deletion/retirement, no order rewrite, no restore operation. 200 LineSummary with new version; 404 absent target, 409 stale version, 400 LINE_ALREADY_RETIRED on current-version already-retired target. Other errors: 401/403, 413/415, 503, 500. Cancel is local and must not call this endpoint. No automatic replay/idempotency-key promise.

5.6 API-PL-06 — Product choices

Arguments: optional q, page, lineId. lineId is current editing line; valid unknown line gives 404. Return only active products, excluding every persisted pair for that line including retired pairs. No lineId in create. SKU/name literal search; sort SKU then UUID, page size 50. Client also excludes unsaved local product IDs. 200 ProductChoicePage, current unit/revision observation; rows are hints, not write authorization. No extension or replacement of existing Product master or GET /api/products contract. Errors: 400, 401/403, 404, 503, 500.

5.7 API-PL-07 — Eligible lines

Arguments: required productId; optional q/page. Unknown product 404. Existing inactive product yields product observation and empty items, not eligible rows. Require active line, product and pair, plus exact confirmed unit/revision match. Filter code/name; stable line ordering, size 50. 200 EligibleLinePage from one read-only snapshot. Selection is not reserved; order save/start rechecks under FOR SHARE locks through commit. Historical assigned lines are displayed by order responses, not smuggled into this eligible list. Errors: 400, 401/403, 404, 503, 500.

6. Existing-order contract extension

Existing API-PO-01–04 routes, numeric version, decimal JSON quantity and product fields stay intact. This new document owns additions; no old approved design is rewritten. Detailed order UI states/compatibility tests belong to step 10.

Context / field	Addition and omission semantics
Create body lineId	Optional nullable UUID; missing/null means no assignment in Draft. Non-null requires eligible exact product/pair
Update body lineId	Optional nullable UUID; omitted preserves saved assignment, explicit null requests clearing. Outside Draft accept only exact unchanged value/null; other change rejected
Detail/list response line	Required nullable object {id, code, name, isActive}; null displays 「未設定」 (Unassigned). Values are current line master, not historical name snapshots
Draft -> InProgress	Require non-null currently eligible assignment even if field omitted or old retired assignment kept
Unchanged historical edit	Keep exact pair/null despite retirement; preserve existing product/quantity/status locks. No backfill for old non-Draft rows
Changed Draft product	Client clears/revalidates line; server rejects incompatible retained non-null pair. No automatic substitution

Draft-origin assignment changes and the transition to InProgress are checked in one transaction. Existing order authorization remains unchanged. Product active and line/pair/unit eligibility checks use DB-approved product -> line -> pair locks, not only FK enforcement. Errors add LINE_REQUIRED (400, lineId), LINE_INELIGIBLE (400, lineId) and LINE_LOCKED (400, lineId); existing quantity, status, product and order-version errors retain their established codes. Existing clients omitting lineId may save Draft/unmodified history; they cannot start a newly transitioned Draft without a valid line. Coordinated feature rollout and write-pause/migration limits remain those of DB; no live deployment performed.

7. Error catalog and precedence

Application errors follow [RFC 9457](#). Return type URI `urn:pmai:problem:production-lines:<suffix>`, fixed English title, matching status, stable code and traceId; optional errors maps JSON paths to code arrays. No detail/body/SQL/stack/observed product values. UI supplies Japanese messages from the existing catalog; codes below are new proposals for review. Malformed body uses errors.body; rows use productChanges[0].minutesPerUnit or products[0].expectedUnitRevision. Do not echo attacker-controlled property names.

Condition	HTTP / code	Type suffix / field / response
Malformed query/body/scalar, overprecision, immutable payload	400 VALIDATION	validation; stable field paths; retain input
New product absent/inactive	400 LINE_PRODUCT_INVALID	product-invalid; affected productId
Duplicate saved association / invalid retired pair operation	400 LINE_PAIR_INVALID	pair-invalid; affected productId/action
Retire already inactive line	400 LINE_ALREADY_RETIRED	already-retired; no state change
Code unique-index race/duplicate	409 LINE_CODE_CONFLICT	code-conflict; code
Parent version stale	409 LINE_STALE	stale; version; guarded reload
Observed product unit/revision differs	409 LINE_UNIT_STALE	unit-stale; expectedUnitRevision; explicit fresh confirmation
Stale saved confirmation and no explicit intent	409 LINE_UNIT_CONFIRMATION_REQUIRED	unit-confirmation; confirmUnit
Line/pair/read product target not found	404 NOT_FOUND	not-found; generic target message
Oversize body / non-JSON mutation	413 REQUEST_TOO_LARGE / 415 UNSUPPORTED_MEDIA_TYPE	request-too-large / unsupported-media-type
Lock timeout/deadlock, rollback confirmed	503 LINE_BUSY	busy; Retry-After: 1; user may retry known failure, no automatic mutation replay
Unexpected/integrity or unverified commit failure	500 UNEXPECTED	internal; generic message, ambiguous write must be verified
Missing session / wrong role	401 / 403, no application code	Established empty body; no login HTML

Authentication/authorization runs first. Shape/bounds next; lock products in UUID order before parent/pairs. After locks, validate target/version before unit/pair business conflicts. Duplicate code at commit still maps named index 23505 to 409. No partial-success/multi-status response. Read and write integrity failures are not converted into invented empty data. Existing order errors remain their own problem type family. Generic middleware fallback may use the existing internal type/code; the feature boundary must not expose its exception detail.

8. Resource deadlines and telemetry

Proposed implementation settings, not currently configured behavior:

Limit / policy	Contract
Feature use-case deadline	15 seconds, linked to request cancellation
DB statement / lock waits	statement_timeout 10s; lock_timeout 5s, transaction-local for this feature; FN defines exact setup and disposal
Read snapshots	Read-only REPEATABLE READ per result, bounded deadline; do not retain across browser pages
Client deadline	20s feature adapter; cancelled/timeout mutation has unknown outcome until verified. Abort does not prove rollback
Retry	Explicit read retry; write retry only after known failure/verification. No global EF retry strategy that replays ambiguous mutations

PostgreSQL distinguishes statement and lock timeouts; use transaction-local configuration, not a blanket change to existing endpoints. See [connection-default documentation](#). Deadlock/timeout 503 requires proven aborted/rolled-back transaction; if commit may have succeeded, do not label it a safely retryable write. FN will detail this.

ActivitySource/Meter name: ProductionManagementAI.ProductionLines. Spans: ProductionLine.List, .Get, .Create, .Update, .Retire, .ProductChoices, .Eligible; FN-036 contributes bounded validation spans to existing order traces. Register source/meter in existing OpenTelemetry pipeline, export only under its existing configuration. Nothing is implemented by this document.

Metrics: pmai.production_lines.requests counter and pmai.production_lines.duration seconds histogram. Labels only operation (list/get/create/update/retire/product_choices/eligible/order_validation) and outcome (success/validation/not_found/forbidden/conflict/busy/error/cancelled/unknown). No raw paths, UUIDs, names, code/search, coefficients, cookies, versions or request bodies in labels. Authorization failures are captured once at feature boundary; known use-case outcomes once after classification, preventing double counting. Structured failure logs carry operation, classified outcome and traceId only. SQL parameter values are not enabled. FN defines exact instrumentation sites.

9. Persistence mapping and verification viewpoints

Contract / requirement	DB mapping / later verification
Identity and hours, REQ-064/065/066	production_lines, code lower index, exact numeric checks; query bounds, duplicates and decimal string binding
Timing actions and unit observation, REQ-066	Composite pair PK, product unit_revision trigger; ABA, stale intent and pair-only parent xmin
Retirement/history, REQ-067	RESTRICT FKs and no runtime DELETE; omission not deletion, Cancel no request, retired pair no re-add

Contract / requirement	DB mapping / later verification
Order assignment, REQ-068	Nullable (line_id, product_id) FK; eligible start/lock serialization, legacy null and omitted-field compatibility
Access/telemetry, REQ-069	Confirmed Admin/Operator read/write roles (DEC-010); per-column grants, no-store, body limits and bounded labels

Later tests must include missing/wrong-typed tokens, "1.2340", "1e1", "0.001", maximum boundaries, multiple changes where one fails, unit change/back-to-original, retirement/start races, new and paged pair reads, route precedence, malicious unknown fields and lost create/update responses. Test IDs are assigned at step 11. No endpoint, SQL, timeout, migration or application test has been executed.

Unresolved business decisions: none; DEC-010 confirms Admin/Operator reads. Technical limits, transport and action/error names are review proposals within approved scope. FN/SPD/order-impact and final family/security consistency remain future work. Submit this finalized version 2 API file and its current EN/JA PDFs; wait for explicit review before authoring DD-FN. No implementation approval inferred.